ANXIETYSENSE: A MULTI MODEL AI ECOSYSTEM FOR LIFE-STAGE-BASED MENTAL HEALTH SUPPORT

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ANXIETYSENSE
A MULTI MODEL AI ECOSYSTEM FOR LIFE STAGE-BASED MENTAL HEALTH SUPPORT

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MULTIMODAL CHILD ANXIETY MANAGEMENT ANALYSIS

SLIIT FACULTY OF COMPUTING

INTRODUCTION TO THE PROBLEM

In Sri Lankan schools, 1 in 4 children struggles with anxiety — most go undetected.

School counselors are overwhelmed. Paper-based questionnaires are easily skipped or faked. Stigma prevents kids from speaking up.

And AI tools? Built on Western data — they fail to read the subtle, cultural expressions of a Sri Lankan child's fear.

So anxiety thrives in silence — harming academic success, social life, and future well-being.

KNOWLEDGE GAP

1. Current AI tools for child anxiety are mostly built on Western datasets, making them unsuitable for other cultures such as Sri Lanka
2. They do not provide age-appropriate or engaging activities for children.
3. No existing system combines the clinically validated SCARED questionnaire with culturally adapted facial expression analysis.
4. Most tools feel too clinical and boring, which causes children to lose interest.

OUR SOLUTIONS

- Develop a machine learning model for questioner specifically trained on Sri Lankan children.
- Combine the SCARED questionnaire with facial expression analysis for more accurate results.
- use explainable AI methods so that the results are clear and trustworthy.
- Provide anxiety-level-based activities to support children effectively.

RESEARCH GAP

Feature	Traditional Clinical Assessment (Birmaher et al., 1999)[1]	Existing AI Emotion Detection (IEEE, 2022)[4]	Digital Mental Health Platforms (General)[6]	Educational Technology (Conventional) [10]	Sri Lankan School Systems (Current)[11,12]	Proposed Anxiety Sense System[This Study]
SCARED Questionnaire Integration	✓	✗	△	✗	△	✓
Facial Emotion Recognition for Children	✗	✓	✗	✗	✗	✓
Cultural Adaptation (Sri Lankan Context)	✗	✗	✗	△	✓	✓
Age-Specific Design (10-16 years)	△	✗	△	✓	△	✓
Gamified Mental Health Assessment	✗	✗	△	✓	✗	✓
Multi-Modal Assessment (ML + Facial)	✗	△	✗	✗	✗	✓
Real-Time Anxiety Level Classification	✗	△	△	✗	✗	✓
Personalized Activity Suggestions	✗	✗	✓	△	△	✓
Privacy-Preserving On-Device Processing	✓	✗	△	△	✓	✓
Weekly Progress Tracking & Points	✗	✗	✓	✓	✗	✓
Crisis Intervention Pathways	✓	✗	△	✗	△	✓

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NOVELTY

Our system is the first designed for Sri Lankan children, combining the SCARED questionnaire with facial emotion analysis for higher accuracy. It uses gamification and age-appropriate activities to keep children engaged, while also providing personalized support based on their anxiety level.

TOOLS AND TECHNOLOGIES

SOFTWARE & FRAMEWORKS

- TensorFlow Lite (Mobile ML deployment)
- OpenCV (Computer vision preprocessing)
- Scikit-learn (Random Forest for questionnaire data)
- YOLO (Real-time face detection)
- React Native (Cross-platform mobile app)
- Firebase (Backend services, analytics)
- AES Encryption (Data security)

ASSESSMENT TOOLS

- SCARED Questionnaire (41-item anxiety assessment)

Alternative considerations:

- PyTorch Mobile instead of TensorFlow Lite (lighter weight)
- SQLite for local data storage (privacy compliance)
- Expo (React Native development platform)

EVALUATION

Validation Method:

Tested against ground-truth clinical assessments by licensed child psychologists using SCARED questionnaire benchmarks.

Target Accuracy:

≥85% accuracy in classifying anxiety levels (Low, Moderate, High) via multimodal fusion of questionnaire + facial analysis.

Usability Validation:

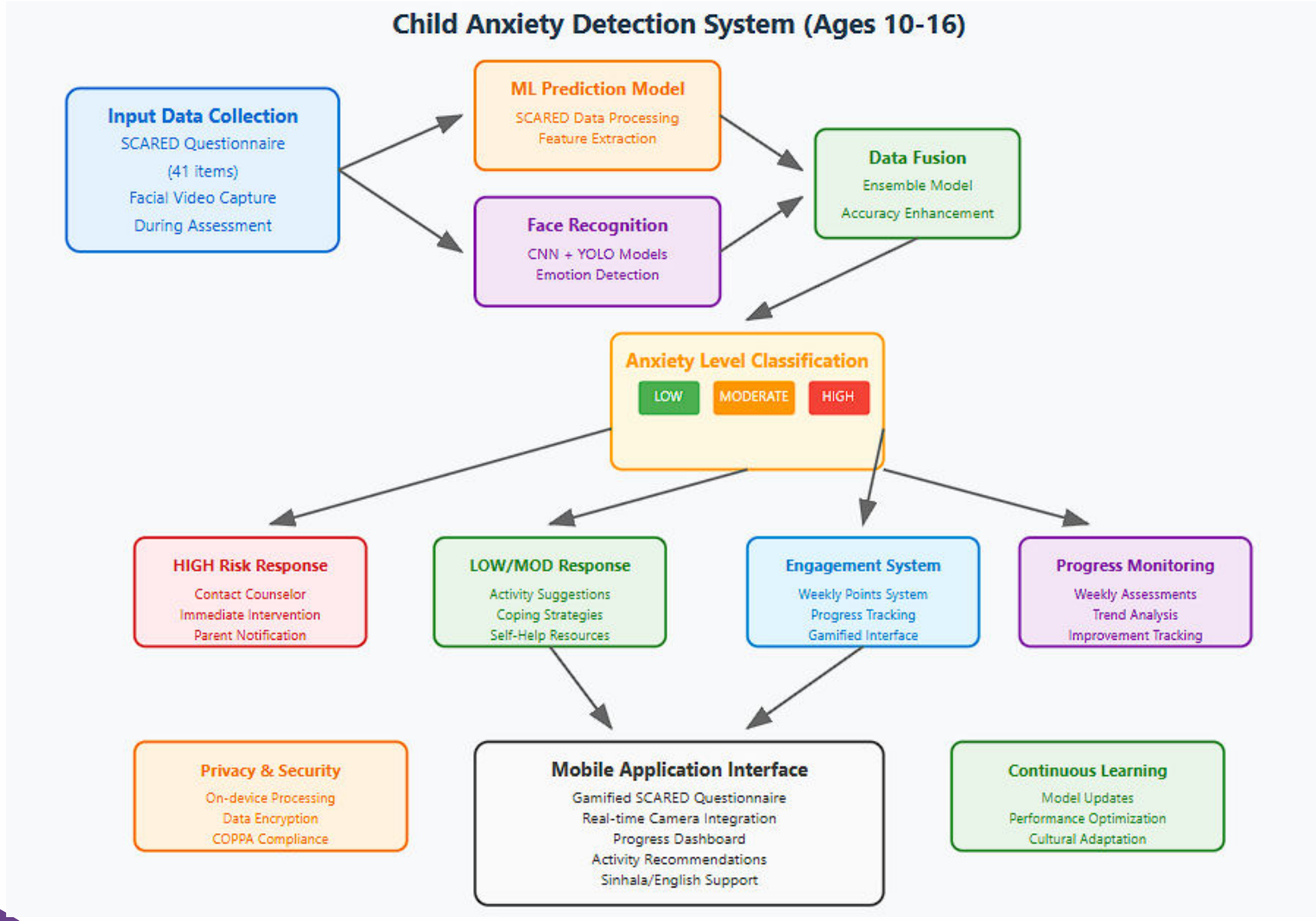
50+ Sri Lankan adolescents (ages 10–16) — tested clarity of Sinhala explanations, gamified engagement, and trust in results.

DATA AVAILABILITY & ETHICAL CLEARANCE

DATA AVAILABILITY & ETHICAL CLEARANCE
(Child Anxiety Detection Component)

- No existing dataset combines SCARED scores + facial expressions from Sri Lankan adolescents.
- Proxy datasets used: FER-2013, RAVDESS (for initial facial emotion model training).
- Primary data collection: Partnering with schools — SCARED responses + facial videos (with parental consent + adolescent assent).
- On-device processing: All facial analysis happens locally — raw video never leaves the phone.
- Ethical clearance: Obtained from SLIIT + school authorities before deployment.
- Culturally adapted: Sinhala language support, age-appropriate design, stigma-sensitive approach.

INDIVIDUAL SYSTEM DIAGRAM



PROJECT REQUIREMENTS

FUNCTIONAL REQUIREMENTS

- Guides children through SCARED questionnaire (41 items)
- Culturally adapted, gamified interface in Sinhala/English.
- Fuses SCARED scores + facial expressions using AI ensemble model
- Combines self-report with CNN based facial emotion recognition.
- Predicts anxiety level: Low, Moderate, High — instantly
- Real-time classification using multimodal fusion.
- Triggers personalized interventions
- Suggests breathing exercises (Low), peer activities (Moderate), or counselor alerts (High)

NON FUNCTIONAL REQUIREMENTS

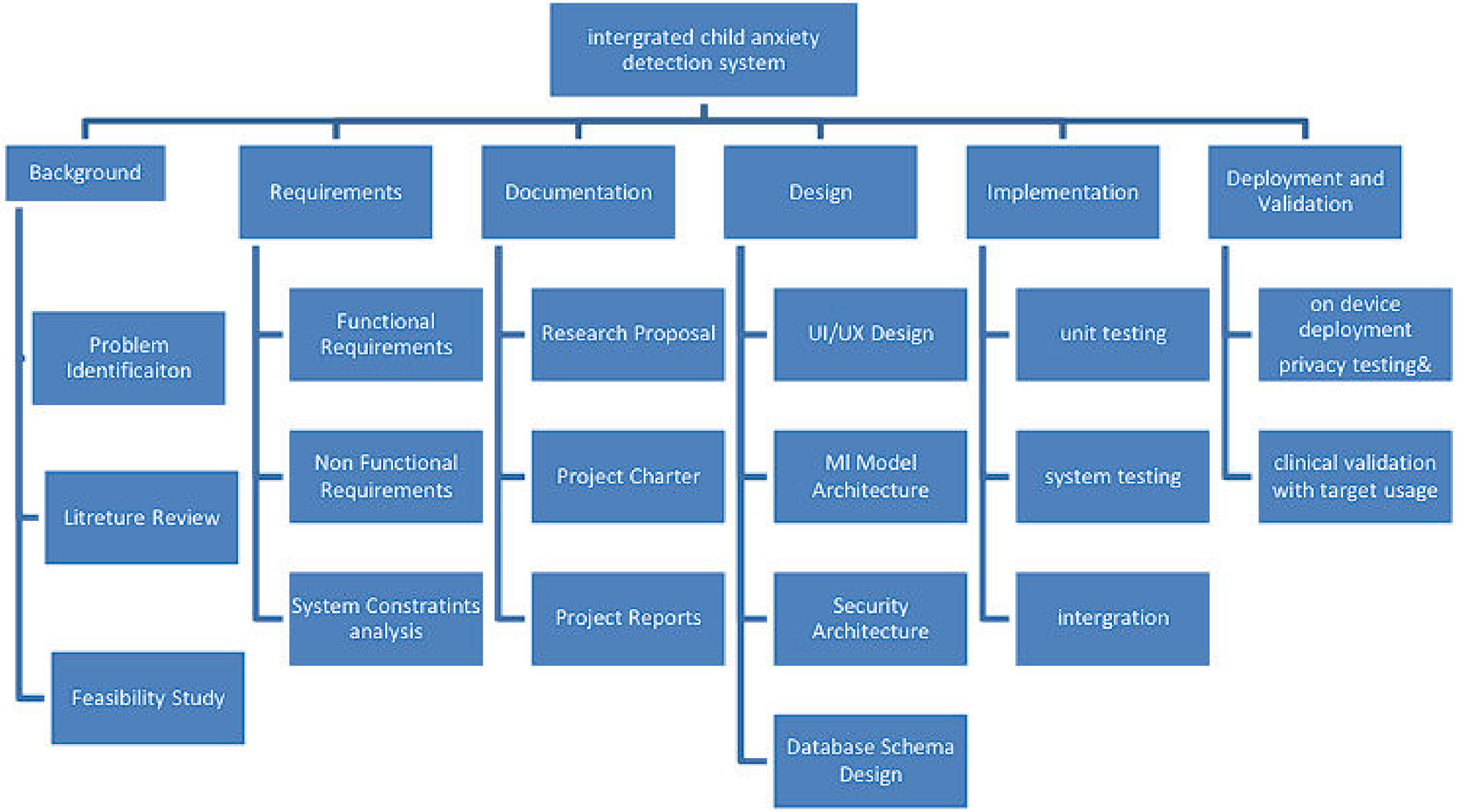
Performance – Prediction delivered in ≤5 seconds (TC-06).

→ Privacy & Security – 100% on-device processing — no raw video leaves phone.

→ Usability & Accessibility – Sinhala-first UI, intuitive for adolescents, supports low-digital-literacy users.

→ Reliability & Scalability – Supports 100+ concurrent users (TC-07).

WORK BREAKDOWN STRUCTURE



REAL WORLD SCENARIO

Meet Samantha, 14, a student from Colombo. Feeling overwhelmed by school pressure but too scared to talk, she opens AnxietySense at home.

In just 10 minutes, she completes the SCARED questionnaire, records a short video of her face while answering, and gets a result: "Moderate Anxiety."

The app explains why — "Your facial cues during questions about school stress + your SCARED score" — in simple Sinhala. A chatbot guides her to breathing exercises and a peer-support group at her school. No stigma. No data leaves her phone.

If her level were "High Risk"? One tap connects her to a school counselor.

AnxietySense doesn't just detect anxiety — it delivers dignity, privacy, and care — exactly where and how Sri Lankan adolescents need it.

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MULTIMODAL YOUNG ADULT ANXIETY MANAGEMENT ANALYSIS

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INTRODUCTION TO THE PROBLEM

In Sri Lanka, many young adults (18–35) struggle with anxiety. About 1 in 4 are affected, but less than 15% get help.

Barriers include shame, Money, culturally unsuitable tools, and too few specialists.

Without treatment, anxiety can cause school dropouts, job loss, substance abuse, and even suicide, which can be prevented.

BACKGROUND

1. GAD-7: accurate but one-time and not culture-friendly.
2. FER: works but needs the cloud (privacy risk) and struggles in low light or with different skin tones.
3. AI models: confusing ‘black boxes’ that users don’t trust.
4. No system adjusts the interface or support in real time based on emotions.”

OUR SOLUTIONS

- GAD-7 Integration and SVM Level prediction
- Real - Time Facial Emotion Recognition
- Triggers Mood-Adaptive UI
- Gamified Interventions

RESEARCH GAP

Feature	Spitzer et al., 2006 [5] (GAD-7 Development)	Löwe et al., 2008 [18] (GAD-7 Validation)	Gao & Al, 2021 [13] (Facial Emotion Recognition)	Fleming et al., 2017 [6] (Gamified Mental Health)	Singh & Dandekar, 2020 [19] (YOLO Emotion Recognition)	Mittelstadt et al., 2016 [7] (Ethics of Algorithms)	Proposed System
GAD-7 Clinical Assessment Integration	✓	✓	✗	✗	✗	✗	✓
Real-time Facial Emotion Recognition	✗	✗	✓	✗	✓	✗	✓
Multimodal Integration (GAD-7 + FER)	✗	✗	✗	✗	✗	✗	✓
Risk-Stratified Intervention Delivery	✗	✗	✗	✓	✗	✗	✓
Privacy-Preserving On-Device Processing	✗	✗	✗	✗	✗	✓	✓
Mood-Adaptive UI Design	✗	✗	✗	✗	✗	✗	✓
Gamified Mental Health Interventions	✗	✗	✗	✓	✗	✗	✓
Cultural Adaptation (Sinhala)	✗	✗	✗	✗	✗	✗	✓
Comprehensive Progress Tracking	✗	✗	✗	✗	✗	✗	✓

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NOVELTY

The novelty of AnxietySense lies not in simply using AI for mental health, but in creating the first culturally grounded, privacy-first, multimodal AI system.

It fuses the clinically validated GAD-7 anxiety scale with real-time, on-device facial expression recognition designed specifically for Sinhalaspeaking young adults.

The system goes further by dynamically adapting its entire user interface to match the user's emotional state.

At the same time, it ensures transparency by explaining its insights and recommendations clearly in the user's native language.

TOOLS AND TECHNOLOGIES

SOFTWARE & FRAMEWORKS

- Lightweight CNN / YOLO
- Super Vector Machine(SVM) for level prediction
- GAD - 7
- Tensor Flow lite
- AES Encryption
- React Native
- Firebase

HARDWARE COMPONENTS

→ User Smartphone (Minimum Spec)

- ✓ Android 7.0+ or iOS 12.0+

→ Development & Testing Devices

- ✓ Mid-range Android phones

→ Clinic/Field Data Collection Devices (Optional)

- ✓ Tablets

→ No External Sensors or Wearables

EVALUATION

Technical Validation:
Validated against GAD-7 assessments by licensed mental health professionals.

Target Accuracy:
≥90%

Clinical Validation:
Tested with 100+ Sri Lankan young adults (18–35) for clarity of Sinhala explanations, trust in results, and engagement with gamified interventions.

DATA AVAILABILITY & ETHICAL CLEARANCE

Data Availability

No public dataset exists. AnxietySense will generate its own primary multimodal data:

✓ GAD-7 responses
✓ On-device facial emotion features (raw images never stored)
✓ Behavioral interaction logs
(Public FER datasets used only for initial model training)

Ethical Clearance

- Not yet obtained (proposal stage)
- Built on ethical pillars: Informed consent, on-device privacy, AI transparency, crisis safety, and Sinhala/Tamil cultural adaptation.
- Formal IRB approval required before user testing.

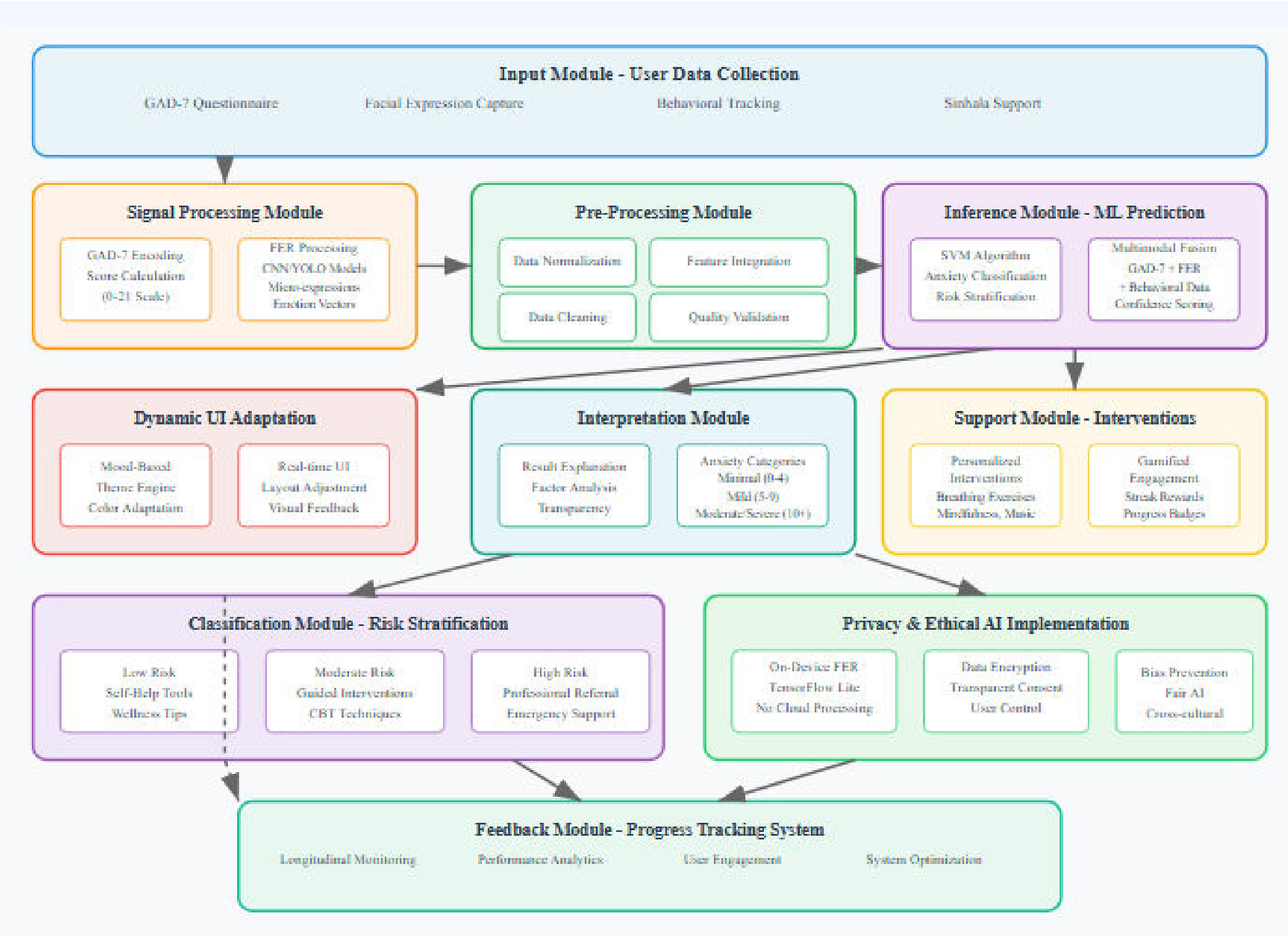
PROPOSED METHODOLOGY

SYSTEM DIAGRAM

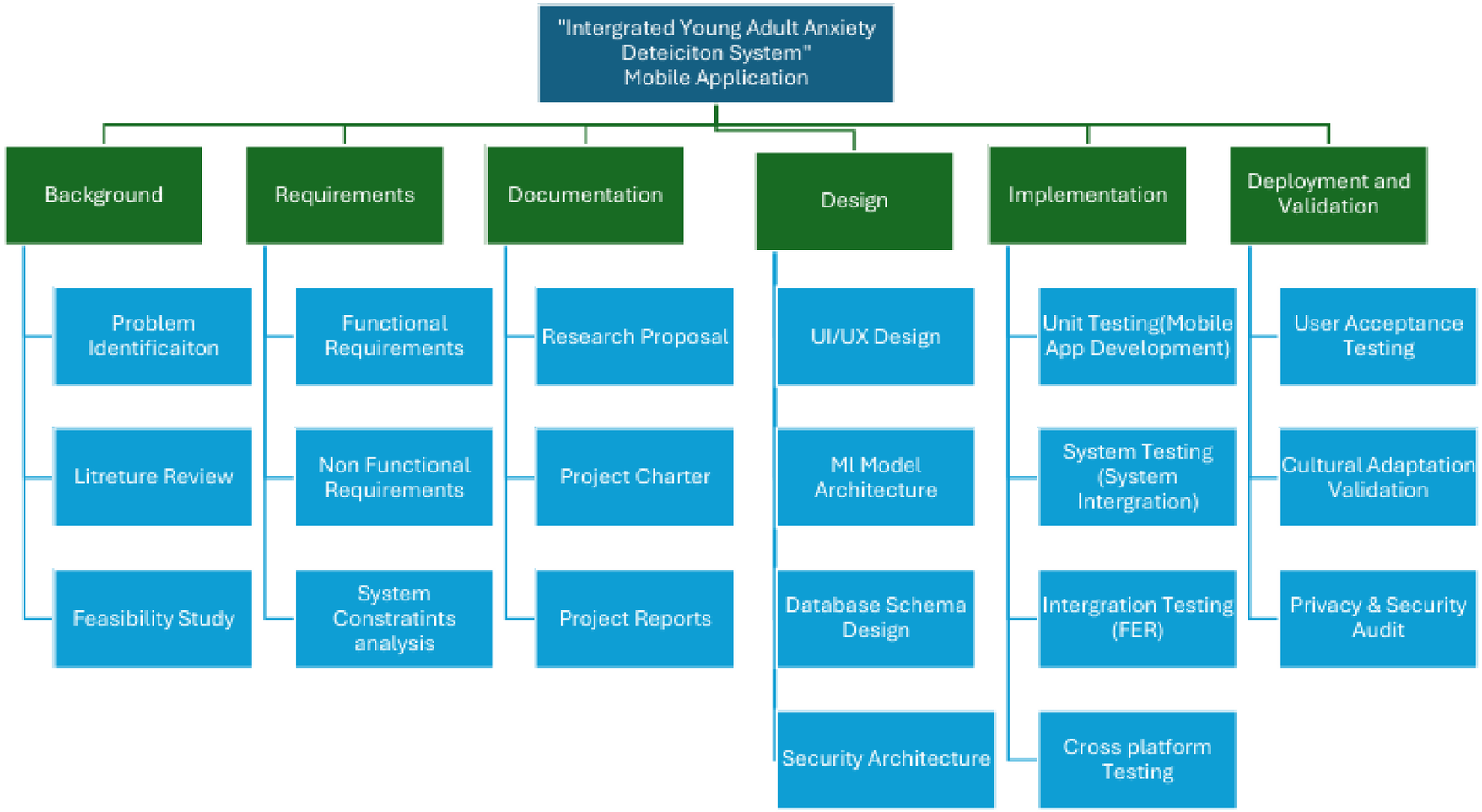
PROJECT REQUIREMENTS

WORK BREAKDOWN STRUCTURE

INDIVIDUAL SYSTEM DIAGRAM



WORK BREAKDOWN STRUCTURE



PROJECT REQUIREMENTS

FUNCTIONAL REQUIREMENTS

- Complete GAD-7 Questionnaire Experience
- Facial Emotion Recognition (FER) Feature
- Smart Anxiety Level Prediction Engine
- Mood-Based UI Themes
- Personalized Intervention Delivery

NON FUNCTIONAL REQUIREMENTS

→ Performance – GAD-7 + FER results in ≤5s.

→ Privacy & Security – 100% on-device; data encrypted.

→ Usability & Accessibility – Sinhala UI; user-friendly; calming themes.

→ Reliability & Accuracy – Accurate integration; works offline.

→ Ethical AI & Explainability – Clear score reasoning; fair across skin tones.

REAL WORLD EXAMPLE

Priya, a university student, feels stressed by exams and family expectations. She opens AnxietySense, is greeted in Sinhala, and completes a consent form ensuring her facial data stays on her phone.

She answers the GAD-7 questionnaire, and a quick on-device facial scan detects severe anxiety, classifying her as high-risk.

The app offers a 2-minute breathing exercise, a gamified “Stress Shield” challenge, and a one-tap connection to a counselor.

After two weeks, her GAD-7 score drops, and the app rewards her progress with UI changes and a “Consistency Badge.”

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PREGNANCY – SPECIFIC ANXIETY DETECTION WITH VOICE ANALYSIS

INTRODUCTION TO THE PROBLEM

In Sri Lanka, 70,000+ pregnant women suffer from anxiety every year — silently.

Clinics are overwhelmed. Paper questionnaires sit unused. Stigma keeps mothers from speaking up.

And AI voice tools? Built for English speakers — they don't understand Sinhala voices, cultural expressions, or pregnancy-specific fears.

So anxiety goes undetected — until it harms both mother and child.

BACKGROUND

1. Current AI voice tools for mental health are not very friendly to Sri Lankan mothers.
2. They're trained on general anxiety — not pregnancy-specific fears expressed in Sinhala
3. No existing system fuses clinically validated PASS questionnaire with culturally adapted voice analysis
4. AI voice tools are trained on Western, English datasets

OUR SOLUTIONS

- Collect & train on voice data from Sinhala-speaking pregnant women.
- Fuse voice biomarkers with PASS scores for hybrid accuracy.
- Use LIME to generate clear explanations accuracy.
- All processing happens on-device. No voice data leaves the phone

RESEARCH GAP

Feature	[1] Chen et al. PASS Validation Study	[6] Asaro et al. Voice Biomarkers	[2] Williams et al. Ensemble Voice Analysis	[3] Johnson et al. Technology Acceptance	[4] Low et al. Speech Analysis Review	[5] Patel & Wong Cultural Adaptation	AnxietySense (Proposed System)
PASS Integration	✓	✗	✗	✓	✗	✗	✓
Voice Analysis	✗	✓	✓	✗	✓	✗	✓
Multimodal Integration (PASS + Voice)	✗	✗	✗	✗	✗	✗	✓
Perinatal-Specific Focus	✓	✗	✗	✓	✗	✗	✓
Sinhala Language Support	✗	✗	✗	✗	✗	✓	✓
On-Device Processing (Privacy Preservation)	✗	✗	✗	✗	✗	✗	✓
Explainable AI (LIME Implementation)	✗	✗	✗	✗	✗	✗	✓
Risk-Stratified Interventions (Low/Moderate/High)	✓	✗	✗	✗	✗	✓	✓

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NOVELTY

The novelty of AnxietySense's voice analysis component lies not in using AI for mental health - but in creating the first culturally grounded, privacy-first, multimodal AI system that fuses clinically validated perinatal anxiety scales with linguistically adapted voice biomarkers - specifically for Sinhala-speaking pregnant women - and explains its decisions transparently in their native language.

TOOLS AND TECHNOLOGIES

SOFTWARE & FRAMEWORKS

- TensorFlow Lite
- Librosa + Parselmouth
- LIME (Local Interpretable Model-agnostic Explanations)
- CatBoost + Gradient Boosting + Logistic Regression
- React Native
- Firebase

HARDWARE COMPONENTS

→ User Smartphone (Minimum Spec)

- ✓ Android 7.0+ or iOS 12.0+

→ Development & Testing Devices

- ✓ Mid-range Android phones

→ Clinic/Field Data Collection Devices (Optional)

- ✓ Tablets

→ No External Sensors or Wearables

EVALUATION

Validation Method:

Tested against ground-truth clinical assessments by licensed perinatal psychologists.

Target Accuracy:

≥85%

Usability Validation:

50+ pregnant women in SriLanka — tested clarity of Sinhala explanations + trust in results

DATA AVAILABILITY & ETHICAL CLEARANCE

No public dataset exists that combines PASS scores with voice recordings from pregnant women — especially in Sinhala. That's why we're partnering with Sri Lankan clinics to ethically collect primary data — making AnxietySense the first of its kind. For initial model validation, we'll use proxy datasets like DAIC-WOZ and OpenSLR — then fine-tune with our own culturally adapted data.

- DAIC-WOZ (voice + anxiety scores)
- OpenSLR (Sinhala speech)

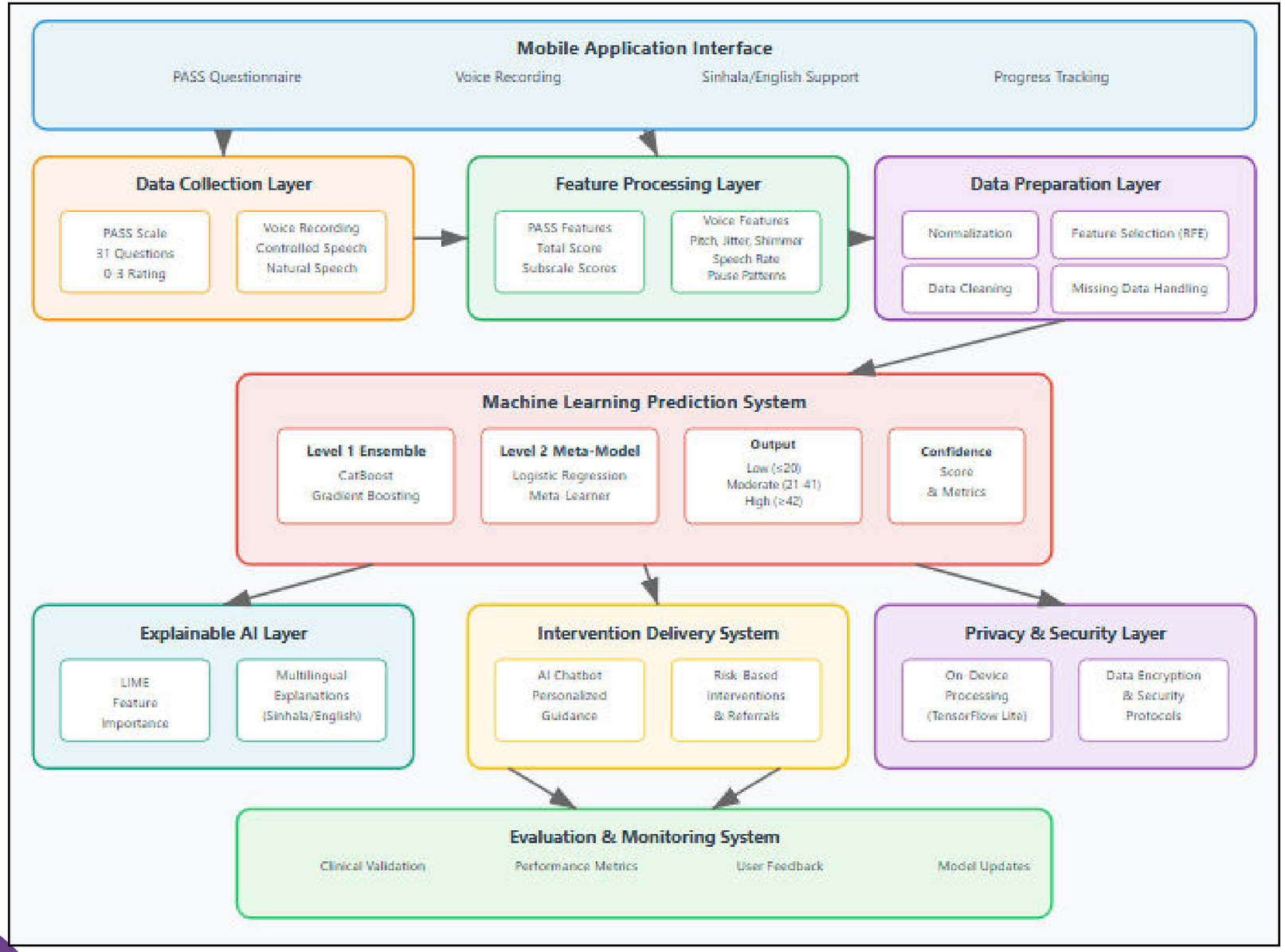
PROPOSED METHODOLOGY

SYSTEM DIAGRAM

PROJECT REQUIREMENTS

WORK BREAKDOWN STRUCTURE

INDIVIDUAL SYSTEM DIAGRAM



PROJECT REQUIREMENTS

FUNCTIONAL REQUIREMENTS

- Guides moms through 31 culturally adapted PASS questions
- Fuses voice + PASS scores using AI ensemble model.
- Predicts anxiety level: Low, Moderate, High — instantly.
- Explains results using LIME, visually clear.
- Triggers chatbot: tailored tips per anxiety risk level

NON FUNCTIONAL REQUIREMENTS

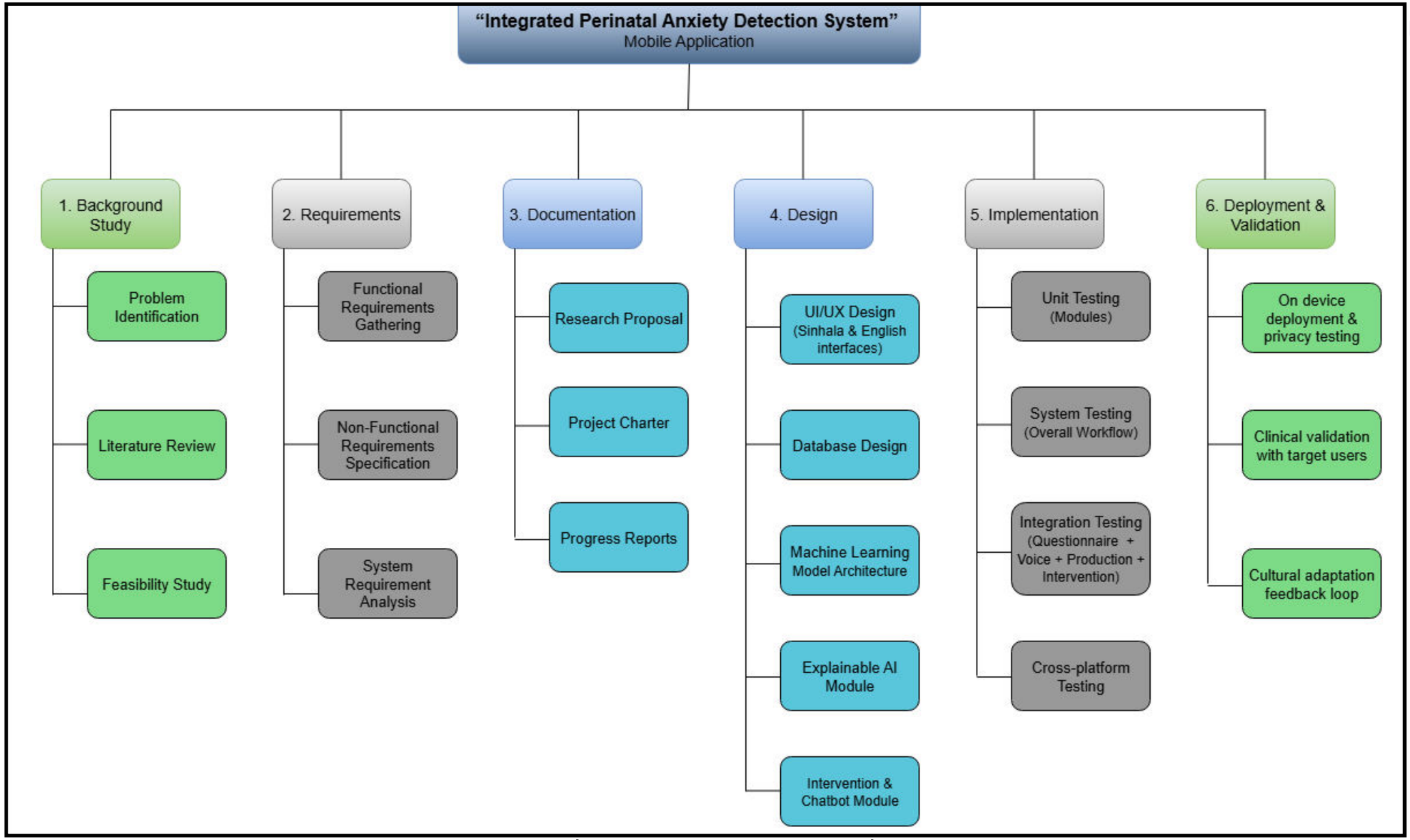
→ Performance - Prediction delivered in ≤5 seconds (TC-05).

→ Privacy & Security - 100% on-device processing — no raw voice leaves phone.

→ Usability & Accessibility - Sinhala-first UI - intuitive for low-digital-literacy users.

→ Reliability & Scalability - Supports 100+ concurrent users (TC-07).

WORK BREAKDOWN STRUCTURE



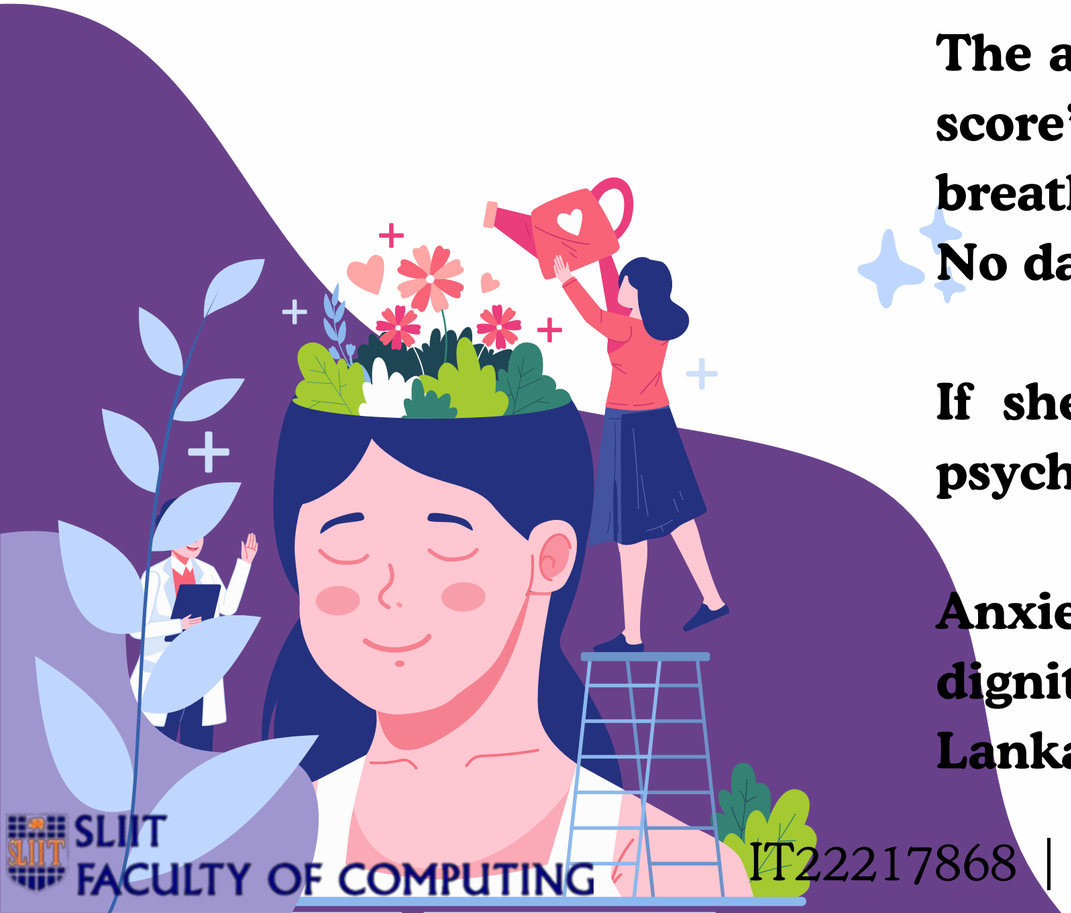
REAL WORLD SCENARIO

Meet Nirmala, 28, pregnant, from Kandy. Overwhelmed but silent - she won't burden her family or wait hours at a clinic. At home, she opens AnxietySense. In 10 minutes she answers 31 questions, records her voice, and gets a result: "Moderate Anxiety."

The app explains why - "Your pauses + fear of childbirth score" - in simple Sinhala. A chatbot guides her to breathing exercises and a local moms' group. No stigma. No data leaves her phone.

If she were "High Risk"? One tap connects her to a psychologist.

AnxietySense doesn't just detect anxiety - it delivers dignity, privacy, and care — exactly where and how Sri Lankan mothers need it.



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ANXIETYSENSE: AI-POWERED, CULTURALLY-ADAPTED MENTAL HEALTH SUPPORT FOR SRI LANKA'S ELDERLY

INTRODUCTION TO THE PROBLEM

Over 300,000 elderly Sri Lankans suffer from undiagnosed anxiety due to stigma and limited access to care. With only 1 psychiatrist per 500,000 people and no geriatric mental health tools in rural clinics, untreated anxiety leads to physical decline and isolation a hidden crisis.

PROBLEM DEFINITION

Sri Lanka has over 300,000 elderly people suffering from anxiety, yet existing digital tools are fragmented, culturally irrelevant, and hard to use. Most rely on Western datasets, lack Sinhala/Tamil support, and raise privacy concerns with cloud-based processing. There is still no integrated, elderly-friendly, and privacy-preserving AI system that combines clinical assessment, emotion recognition, and therapeutic interventions.

BACKGROUND

1

Limited Cultural Adaptation

- Generic playlists without cultural context
- No nostalgic relevance for elderly users

2

No Emotional Integration

- Static recommendations unaware of real-time emotional state
- Missing therapeutic timing

3

Poor Elderly Accessibility

- Complex interfaces not designed for age-related needs

4

Privacy Concerns

- Cloud-based processing violating trust
- Inappropriate for vulnerable populations

OUR SOLUTIONS

- GAS Integration
- Elderly-Optimized Facial AI
- 100% On-Device Processing
- Collaborative Filtering
- Sinhala and English music therapy with songs from their younger days

RESEARCH GAP

Feature	Traditional Geriatric Anxiety Assessment (Clinical Psychology, 2021)	AI-Based Elderly Emotion Detection (IEEE, 2022)	Digital Mental Health Platforms (General)	Music Therapy Systems (Conventional)	Explainable AI in Healthcare	Proposed AnxietySense System
Real-time GAS Assessment Integration	✓	✗	△	✗	✗	✓
Facial Emotion Recognition for Elderly	✗	✓	✗	✗	✗	✓
Cultural Adaptation (Sri Lankan Context)	✗	✗	✗	△	✗	✓
Nostalgic Music Therapy Integration	✗	✗	✗	✓	✗	✓
AI-Powered Personalization	✗	✓	△	✗	✓	✓
Multi-Modal Assessment (GAS + Emotion)	✗	✗	✗	✗	✗	✓
Elderly-Friendly Interface Design	△	✗	✗	△	✗	✓
On-Device Privacy Protection	✓	✗	△	✓	✗	✓
Explainable AI for Elderly Users	✗	✗	✗	✗	△	✓
Bilingual Support (Sinhala/English)	✗	✗	✗	✗	✗	✓
Real-Time Therapeutic Response	✗	✗	△	✗	✗	✓
Birth Decade-Based Personalization	✗	✗	✗	✗	✗	✓

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NOVELTY

The novelty of our system is combining clinically validated geriatric anxiety assessment, elderly-specific facial emotion recognition, and nostalgic music from their youth into a single offline AI platform, designed for accessibility, privacy, and cultural relevance for seniors with low digital literacy.

TOOLS AND TECHNOLOGIES

SOFTWARE & FRAMEWORKS

- TensorFlow Lite → secure on-device AI
- SQLite → offline-first storage & sync
- Collaborative Filtering → nostalgic Sinhala music from the user's youth period

HARDWARE COMPONENTS

→ User Smartphone (Minimum Spec)

- ✓ Android 7.0+ or iOS 12.0+

→ Development & Testing Devices

- ✓ Mid-range Android phones

→ Clinic/Field Data Collection Devices (Optional)

- ✓ Tablets

→ No External Sensors or Wearables

EVALUATION

Validation Method:

- Accuracy vs. clinical results (≥85%)
- Performance (≤5s, 100 users)
- Security (encryption, compliance)
- User testing with elderly

Target Accuracy:

GAS completion ≥90%

Usability Validation:

- Elder-friendly UI (fonts, voice)
- Accessibility (Sinhala, WCAG)

DATA AVAILABILITY & ETHICAL CLEARANCE

- No existing dataset combines GAS scores + facial expressions + nostalgic music preferences for elderly Sri Lankans.
- Partnering with local elder-care homes & clinics to ethically collect primary data.
- For initial model training, we'll use proxy datasets:
  - FER-2013 & CK+ → facial emotion recognition
  - Million Song Dataset → music features (then culturally adapt with Sinhala songs)
- Fine-tuning done on our own culturally adapted elderly dataset.
- Ethical approval sought through SLIIT Ethics Review Committee before data collection.
- Aligns with GDPR + Sri Lankan Health Data Policy.

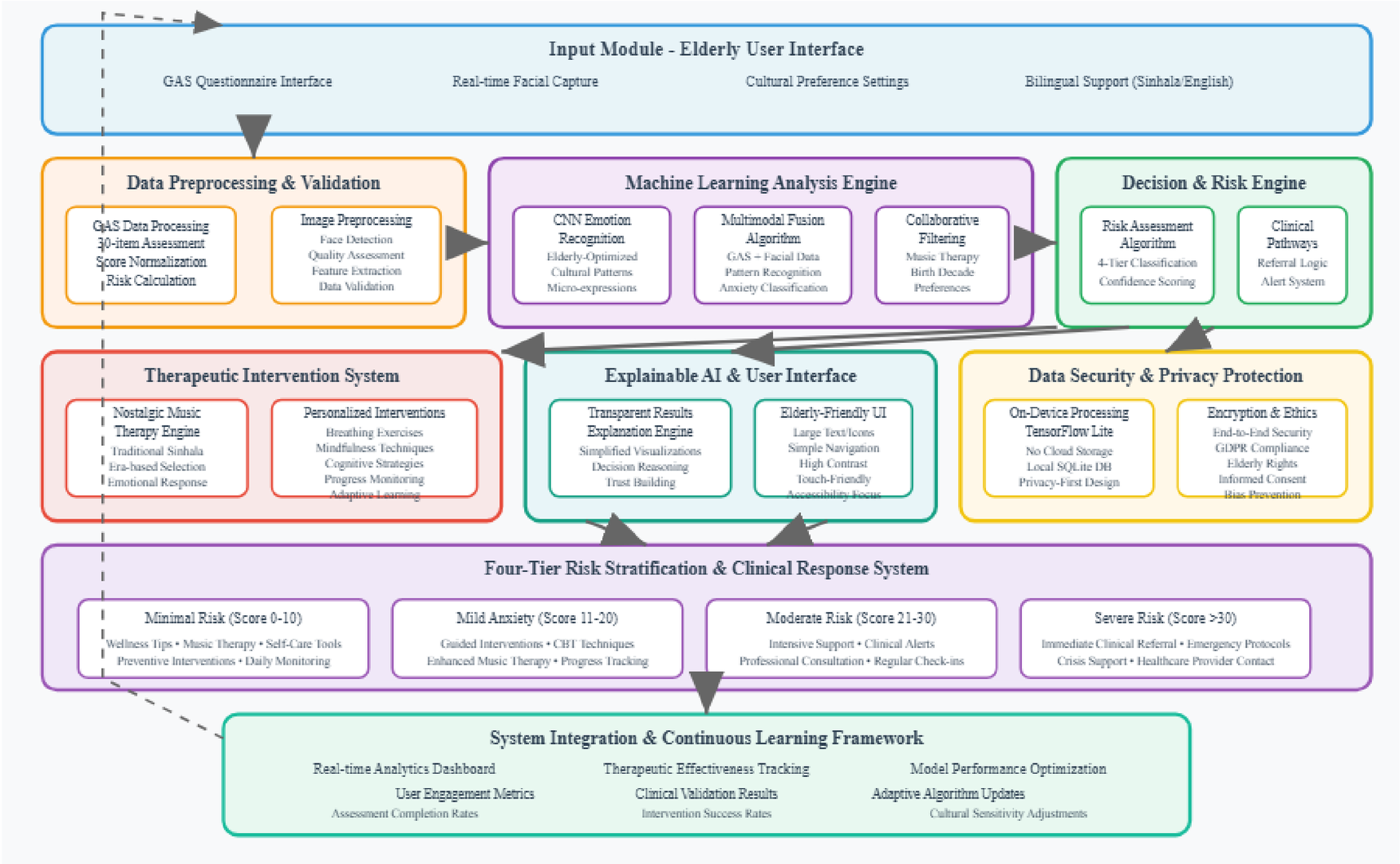
PROPOSED METHODOLOGY

SYSTEM DIAGRAM

PROJECT REQUIREMENTS

WORK BREAKDOWN STRUCTURE

INDIVIDUAL SYSTEM DIAGRAM



PROJECT REQUIREMENTS

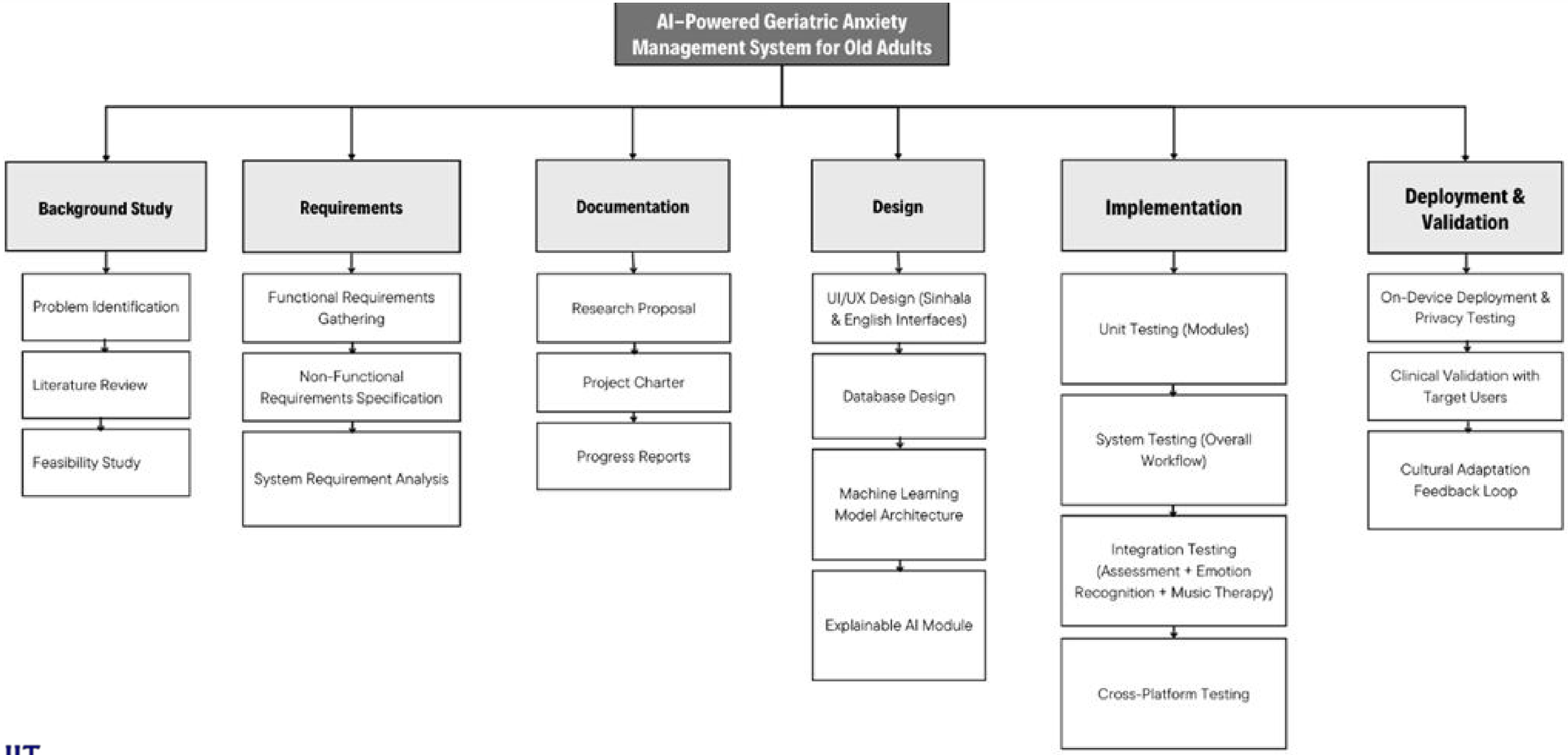
FUNCTIONAL REQUIREMENTS

- GAS questionnaire with elderly-friendly UI (Sinhala/English, voice, large fonts).
- Real-time facial emotion recognition (optimized for elderly).\
- Personalized nostalgic music therapy (birth decade & preferences).
- Smart analysis engine (fast results <5s).
- Four care levels (minimal → severe with clinical referral).

NON FUNCTIONAL REQUIREMENTS

- Fast & accurate (≤5s, ≥85% accuracy).
- Scalable & reliable (99% uptime, backups).
- Secure & private (encryption, on-device processing, GDPR compliant).
- Accessible (bilingual, WCAG compliant, elderly-friendly).
- Maintainable (modular design, easy AI updates).

WORK BREAKDOWN STRUCTURE



REAL WORLD EXAMPLE

Mr. Silva, 68, from Galle, often feels restless and anxious after retiring. He opens the AnxietySense app on his phone. A friendly voice-guided Sinhala questionnaire begins, and the camera quietly observes his subtle facial expressions. The AI detects signs of moderate anxiety. Instead of giving generic advice, the system plays Sinhala songs from the 1980s — his youth period — the music he grew up with. Listening brings him comfort and a sense of familiarity. The app logs his mood improvement for future tracking.

THANK YOU

Feel free to share, support and help each other!